A critical-point criterion for D-stability in every dimension

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Abstract

A real square matrix A is D-stable if DA is Hurwitz for every positive diagonal matrix D. Johnson's contact criterion is expressed as an explicit principal-minor critical-point system. For every dimension and every choice of positive barrier weights, D-stability is equivalent to Hurwitz stability and the absence of positive solutions of the associated Fritz–John equations. A proper exhaustion of the positive orthant ensures that every nonempty contact set contains a critical point. In dimension five, the criterion is stated explicitly as a Lagrange system and a tangency system. Both equivalences are formalized in Lean 4/Mathlib. The paper also develops the geometry of regular and tangential contacts, a criterion for stability under bounded time-scale ratios, and attainment of the minimal destabilizing ratio. These results are accompanied by formal proofs, exact examples and certified numerical calculations, with the evidence for each statement identified individually. A finite decision procedure is discussed separately: generic finiteness is given as a proof sketch, termination is treated conventionally, and exact execution is demonstrated for dimension three. For dimension five the Lagrange system has multihomogeneous Bézout number 1200 and a numerically observed generic count of 980 complex solutions. Applications include a five-species Lotka–Volterra community and a published refinery-column gain matrix.

1 Introduction

A real $n \times n$ matrix A is *D-stable* if DA is Hurwitz (all eigenvalues in the open left half-plane) for every diagonal matrix D with positive diagonal. The notion goes back to Arrow and McManus [2]; it is natural whenever a linear or linearized system is known up to unknown positive rates. Table 1 lists three settings used below.

Table 1: Where diagonal scalings come from. In each row "A is D-stable" means stability for every choice of the positive quantities in the third column.

Setting	Model	Scaling D	D-stability means
Lotka–Volterra communities	$\dot{x}_i = x_i(r_i + (Ax)_i)$	equilibrium abundances x^*	every feasible coexistence equilibrium is linearly stable (Jacobian $\operatorname{diag}(x^*)A$; every $x^* > 0$ is feasible for $r = -Ax^*$)
Process units	$\dot{x}_i = f_i(x)/V_i$	inverse hold-ups $1/V_i$	steady state stable for all hold-ups
Decentralized integral control	loops k_i/s on $G(s)$	loop gains k_i	low-gain stability for all gains; see Section 7.2

Status of the problem. D-stability of 2×2 matrices is equivalent to $-A$ being a P_0^+-matrix [17]. Cain [7] characterized the 3×3 case in closed form with radicals. Kanovei and Logofet [20] gave a 4×4 criterion through local extrema of a coordinate on a Hurwitz hypersurface, and Impram, Johnson and Pavani [16] studied robust D-stability for $n = 4$. The surveys [22, 11] list no complete explicit criterion for $n \geq 5$; Burlakova's conditions for $n = 5$ [6] are described in the literature as partial. Several equivalent Johnson-type reformulations are collected in [24]. D-stability is decidable for every fixed n by quantifier elimination: the D-stable matrices form the complement of a semialgebraic set [19]. Kushel [23] develops recursive determinantal tests and principal-minor sufficient conditions, while describing general characterization above dimension four as open. Pavani [26] also gives equivalent conditions designed for symbolic computation above dimension four. A statement of the form "A is D-stable iff an explicit polynomial system has no positive solution" is therefore available in principle for every n; the criterion below is of the same kind as the 4×4 criterion of Kanovei and Logofet, with a specific square system, stated explicitly for $n = 5$ and checked by machine. The interior of the D-stable set was described by Hartfiel [13] and Abed [1] (strong D-stability); see also Cross [9]. Deciding D-stability is coNP-hard [34]. No proof in this paper depends on the companion manuscripts [33, 34]; they supply this hardness result as context and two test families (Example 6.7 and the coNP gadgets of Section 8).

The method is classical. The starting point is Johnson's criterion [18]: a Hurwitz matrix is D-stable iff $\det(iI - A \operatorname{diag} d)$ has no zero with $d > 0$ (Lemma 2.3). To certify emptiness of this real set we use the foundational lemma of the *critical point method* [12, 27, 28, 29, 4, 3, 30]: a proper function attains its minimum on a nonempty closed set, and the minimizer is a Fritz–John point. Degree bounds for critical-point systems of the multihomogeneous kind below are in [31], and the handling of open orthants by an inequation in [5]. Our contribution is the application of this method to Johnson's contact set, its formal proof, and the consequences below, not the method itself.

Results and their status. We label every statement: *Lean* (formalized in Lean 4 with Mathlib, source verification with warnings treated as errors; Section 10), *exact* (exact computation), *conventional* (a complete written proof, not formalized), *sketch* (a proof outline), or *numerical*.

(i) *Criterion, every n (Lean).* A is D-stable iff it is Hurwitz and the Fritz–John system of a proper exhaustion on the contact set has no positive solution (Theorem 3.2); in principal-minor coordinates it splits into a Lagrange branch and a tangency branch (Theorem 3.4, Corollary 3.5), stated with explicit coefficients for $n = 5$ (Theorem 3.6). These are equivalences, not decision procedures. The proper exhaustion handles the noncompact positive orthant directly.

(ii) *The equality branch (Lean and conventional).* Regular contacts persist under perturbation (Theorem 4.3); hence a Hurwitz matrix in the closure of the D-stable set is either D-stable without contact or not D-stable with only tangential contacts (Theorem 4.5). An exact example shows that the tangency branch cannot be dropped (Example 4.6). A finite recursion decides the tangency branch without any finiteness assumption (Lemma 4.8, conventional).

(iii) *Size (exact and sketch).* The Lagrange system has multihomogeneous Bézout number $n!\binom{n}{2}$ (Proposition 5.1) and is finite and nonsingular for generic weights (Proposition 5.2, sketch). The resulting finite procedure was executed exactly for $n = 3$ only.

(iv) *Time-scale robustness (Lean and conventional).* Stability for all rate vectors with pairwise ratios at most κ is decided by Fritz–John points on the faces of a compact box

(Theorem 6.2); the minimal destabilizing ratio κ^* is attained (Proposition 6.3); near a Hurwitz matrix without contacts destabilization needs ratios beyond any bound (Theorem 6.4), at a rate $\kappa^* \asymp \delta^{-1}$ that is only sketched (Remark 6.6) and observed numerically; near nondegenerate tangential points it does not (Proposition 6.5, conventional).

2 Setting

For $d \in \mathbb{R}^n$ write $\operatorname{diag} d$ for the diagonal matrix with entries d_i, and $\mathbb{R}^n_{>0}$ for the open positive orthant. Since $A \operatorname{diag} d$ and $\operatorname{diag}(d)A$ are similar for $d > 0$, D-stability can be tested with right scalings; we do so throughout, as the formalization does. For $S \subseteq \{1, \dots, n\}$ let $m_S = \det(-A[S])$ be the principal minor of $-A$ on S, with $m_\emptyset = 1$, and $d^S = \prod_{l \in S} d_l$.

Definition 2.1. The *contact determinant* of A is $f_A(d) = \det(iI - A \operatorname{diag} d)$ for $d \in \mathbb{R}^n$, with real and imaginary parts $P_{\mathrm{re}}, P_{\mathrm{im}}$. A *contact* is a point $d \in \mathbb{R}^n_{>0}$ with $f_A(d) = 0$; the (semialgebraic) set of contacts is the *contact set* C_A. Its complexification is $V_A = \{d \in \mathbb{C}^n : P_{\mathrm{re}}(d) = P_{\mathrm{im}}(d) = 0\}$, of codimension two.

A contact is a positive rate vector at which $A \operatorname{diag} d$ has the eigenvalue i. If $A \operatorname{diag} d$ has an eigenvalue $i\omega$ with $\omega > 0$, then d/ω is a contact. Note that $|f_A(d)|^2 = \det(I + (A \operatorname{diag} d)^2)$, the quantity used in [13, 1].

Lemma 2.2 (Principal-minor expansion; Lean)**.** *$f_A(d) = \sum_{S \subseteq \{1,\dots,n\}} i^{\,n-|S|}\, m_S\, d^S$. In particular P_{re} and P_{im} are multiaffine polynomials whose coefficients are principal minors of $-A$.*

Proof. The l-th row of $(iI - A \operatorname{diag} d)^{\mathsf{T}}$ is d_l times column l of $-A$ plus i times the unit row e_l. Expanding the determinant multilinearly in the rows gives, for each S, the factor $i^{n-|S|} d^S$ times the determinant of the matrix whose rows are the columns of $-A$ indexed by S and unit rows elsewhere, which equals $\det(-A[S])$ by a block-triangular expansion. □

Lemma 2.3 (Johnson's criterion [18]; Lean)**.** *A is D-stable if and only if A is Hurwitz and $C_A = \emptyset$.*

Proof. A contact is an eigenvalue i of $A \operatorname{diag} d$, so D-stable matrices have none. Conversely, let A be Hurwitz without contacts. No $A \operatorname{diag} d$ with $d > 0$ has an eigenvalue on the imaginary axis: the eigenvalue 0 is excluded because A is nonsingular, and $i\omega$ with $\omega \neq 0$ would give the contact $d/|\omega|$ (use the complex conjugate if $\omega < 0$). The orthant is connected, the Hurwitz matrices and the matrices with an eigenvalue of positive real part are open, and $A \operatorname{diag} \mathbf{1} = A$ is Hurwitz; hence every $A \operatorname{diag} d$ is Hurwitz. □

3 Coverage and the explicit criterion

Definition 3.1. A *proper exhaustion* of $\mathbb{R}^n_{>0}$ is a function $\psi\colon \mathbb{R}^n_{>0} \to \mathbb{R}$ of class C^1 each of whose sublevel sets $\{d \in \mathbb{R}^n_{>0} : \psi(d) \le r\}$ is contained in a compact subset of $\mathbb{R}^n_{>0}$. A *Fritz–John point* of ψ on C_A is a contact d for which there is $(\Lambda_1, \Lambda_2, \Lambda_0) \neq 0$ with $\Lambda_1 \nabla P_{\mathrm{re}}(d) + \Lambda_2 \nabla P_{\mathrm{im}}(d) + \Lambda_0 \nabla \psi(d) = 0$.

The weighted log barrier $\psi_{a,b}(d) = \sum_i (a_i d_i - b_i \log d_i)$ with $a, b > 0$ is a proper exhaustion, and so is $\sum_i (a_i d_i + b_i/d_i)$.

Theorem 3.2 (Coverage; Lean)**.** *Let ψ be a proper exhaustion of $\mathbb{R}^n_{>0}$. A real square matrix A is D-stable if and only if A is Hurwitz and ψ has no Fritz–John point on C_A.*

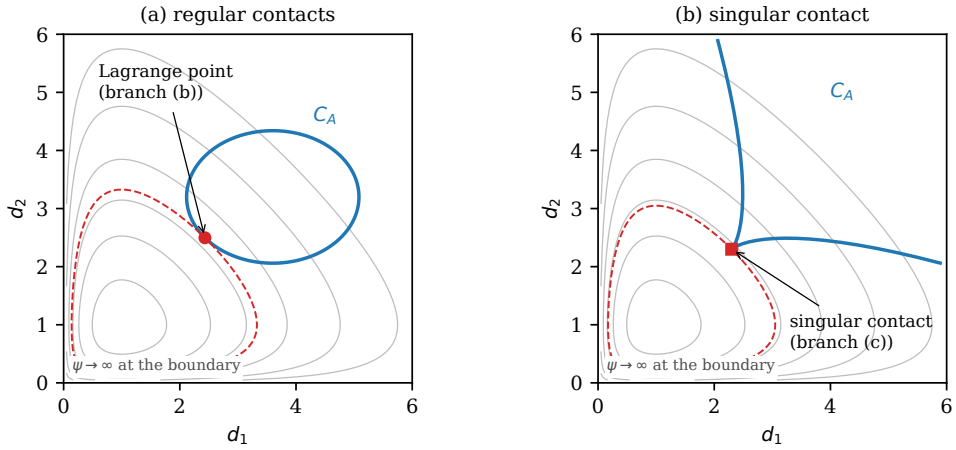


Figure 1: Schematic of the coverage argument, drawn in two dimensions (the contact set itself has dimension $n-2$). Grey: level sets of $\psi(d) = \sum_i (d_i - \log d_i)$, which blow up at the boundary of the orthant. Blue: contacts. The minimizer of ψ over the contacts is either a regular point where a level set touches C_A (a solution of (b), panel (a)) or a singular point of C_A (a solution of (c), panel (b)).

Proof. If A is D-stable it is Hurwitz and $C_A = \emptyset$ (Lemma 2.3), so there is no Fritz–John point. Conversely let A be Hurwitz and not D-stable. By Lemma 2.3 there is a contact d_0. The set $K = \{\psi \le \psi(d_0)\} \cap C_A$ is compact (a closed subset of a compact sublevel set) and contains d_0, so $\psi|_K$ attains its minimum at some d^*. Contacts outside K have $\psi > \psi(d_0) \ge \psi(d^*)$, and $\mathbb{R}^n_{>0}$ is open, so d^* is a local minimum of ψ on $\{P_{\mathrm{re}} = 0,\ P_{\mathrm{im}} = 0\}$. The Lagrange multiplier rule in Fritz–John form (no constraint qualification) gives $(\Lambda_1, \Lambda_2, \Lambda_0) \neq 0$. □

Remark 3.3 (Relation to the 4×4 criterion of Kanovei and Logofet)**.** Kanovei and Logofet [20, Thm. 5] reduce the 4×4 case to a slice condition at $z = 1$ and the absence of positive local extrema of the coordinate z on $T = \{f = 0\} \subset \mathbb{R}^3_{>0}$, where f is the Hurwitz determinant of $\mathrm{diag}(1, x, y, z)A$; besides contacts, T contains points where the scaled matrix has a pair of real eigenvalues $\pm r$. Their argument distinguishes whether T is bounded away from $z = 0$ and $z = \infty$. Two points are not addressed explicitly there: z is not proper on T, which is closed only in the open orthant, so extremizing sequences may escape to the boundary in x or y while z stays bounded; and the dichotomy uses connectedness of $z(T)$. We do not claim a counterexample to their criterion. With a proper exhaustion neither point arises, and Theorem 3.2 with $n = 4$ is a complete criterion.

In principal-minor coordinates the Fritz–John condition is explicit. Write $\partial_i f_A(d) = \sum_{S \ni i} i^{\,n-|S|} m_S\, d^{S\setminus\{i\}}$.

Theorem 3.4 (Explicit criterion; Lean)**.** *Let $a, b \in \mathbb{R}^n_{>0}$. A is D-stable if and only if A is Hurwitz and there is no $d \in \mathbb{R}^n_{>0}$ with $f_A(d) = 0$ for which some $(\Lambda_1, \Lambda_2, \Lambda_0) \neq 0$ satisfies, for every i,*

$$\Lambda_1 \operatorname{Re} \partial_i f_A(d) + \Lambda_2 \operatorname{Im} \partial_i f_A(d) + \Lambda_0 \Big(a_i - \frac{b_i}{d_i}\Big) = 0.$$

Proof. Theorem 3.2 for $\psi_{a,b}$, with the partial derivatives of Lemma 2.2 evaluated on the unit vectors. □

Corollary 3.5 (Two branches; Lean)**.** *A is D-stable if and only if A is Hurwitz and neither of the following systems has a solution with $d \in \mathbb{R}^n_{>0}$:*

(b) *(Lagrange branch)* $f_A(d) = 0$ *and, for some* $\lambda, \mu \in \mathbb{R}$ *and all* i, $a_i d_i - b_i = \lambda\, d_i \operatorname{Re} \partial_i f_A(d) + \mu\, d_i \operatorname{Im} \partial_i f_A(d)$;

(c) *(tangency branch)* $f_A(d) = 0$ *and, for some* $t \in \mathbb{R}$ *and all* i, $(1-t^2) \operatorname{Re} \partial_i f_A(d) + 2t \operatorname{Im} \partial_i f_A(d) = 0$.

Proof. If $\Lambda_0 \neq 0$, divide by Λ_0 and multiply the i-th equation by d_i: this is (b) with $\lambda = -\Lambda_1/\Lambda_0$, $\mu = -\Lambda_2/\Lambda_0$. If $\Lambda_0 = 0$ then $(\Lambda_1, \Lambda_2) \neq 0$, and every nonzero direction in $\mathbb{R}^2$ is, up to a nonzero factor, $(1-t^2, 2t)$ for a real t: with $r = (\Lambda_1^2 + \Lambda_2^2)^{1/2}$ take $t = 0$ if $\Lambda_2 = 0$ and $t = (r - \Lambda_1)/\Lambda_2$ otherwise. Conversely (b) gives $(\Lambda_1, \Lambda_2, \Lambda_0) = (-\lambda, -\mu, 1)$ and (c) gives $(1-t^2, 2t, 0) \neq 0$. □

For $n = 5$ set $c_k(d) = \sum_{|S|=k} m_S d^S$ ($c_0 = 1$) and $c_k^{(i)}(d) = \sum_{|S|=k,\, S \ni i} m_S\, d^{S \setminus \{i\}}$. Since $i^{5-k} = i, 1, -i, -1, i, 1$ for $k = 0, \dots, 5$, Lemma 2.2 gives $P_{\mathrm{re}} = c_1 - c_3 + c_5$ and $P_{\mathrm{im}} = 1 - c_2 + c_4$.

Theorem 3.6 (The 5×5 criterion; Lean)**.** *Let* A *be a real* 5×5 *matrix and* $a, b \in \mathbb{R}^5_{>0}$. *Then* A *is D-stable if and only if*

(a) A *is Hurwitz;*

(b) *there are no* $d \in \mathbb{R}^5_{>0}$ *and* $\lambda, \mu \in \mathbb{R}$ *with* $c_1 - c_3 + c_5 = 0$, $1 - c_2 + c_4 = 0$ *and*

$$a_i d_i - b_i = \lambda\, d_i (c_1^{(i)} - c_3^{(i)} + c_5^{(i)}) + \mu\, d_i (c_4^{(i)} - c_2^{(i)}) \quad (i = 1, \dots, 5);$$

(c) *there are no* $d \in \mathbb{R}^5_{>0}$ *and* $t \in \mathbb{R}$ *with* $c_1 - c_3 + c_5 = 0$, $1 - c_2 + c_4 = 0$ *and*

$$(1-t^2)(c_1^{(i)} - c_3^{(i)} + c_5^{(i)}) + 2t(c_4^{(i)} - c_2^{(i)}) = 0 \quad (i = 1, \dots, 5).$$

System (b) is square: seven equations in (d, λ, μ). *System (c) has seven equations in six unknowns.*

Remark 3.7 ($n = 3$ and $n = 4$; conventional)**.** For $n = 3$, $f_A = (c_3 - c_1) + i(c_2 - 1)$. Assume $-A \in P_0^+$, which D-stability requires; then $c_2 > 0$ on the orthant, and by homogeneity ($c_k(sd) = s^k c_k(d)$) a contact exists iff $\Delta_2 = c_1 c_2 - c_3$ has a positive zero. Now $\Delta_2/(d_1 d_2 d_3) = (\sum m_i d_i)(\sum m_{jk}/d_i) - m_{123}$, and by the Cauchy–Schwarz inequality the first product has infimum α^2 with $\alpha = \sqrt{m_1 m_{23}} + \sqrt{m_2 m_{13}} + \sqrt{m_3 m_{12}}$, attained (at $d_i \propto \sqrt{m_{jk}/m_i}$) iff for every i the minors m_i and m_{jk} vanish together. Hence a Hurwitz A with $-A \in P_0^+$ is D-stable iff $m_{123} < \alpha^2$ in the attained case and $m_{123} \leq \alpha^2$ otherwise, which is Cain's criterion [7]; in the attained case with $m_{123} = \alpha^2$ the unique contact is the Cauchy–Schwarz point, and it is singular (Example 4.6). For $n = 4$, Corollary 3.5 is a complete criterion by a square system in $(d_1, \dots, d_4, \lambda, \mu)$ with Bézout number 144 and a tangency system of six equations in five unknowns.

4 The equality branch

Definition 4.1. A contact d is *regular* if the real derivative $Df_A(d) \colon \mathbb{R}^n \to \mathbb{C} \cong \mathbb{R}^2$ is onto, i.e. $\operatorname{rank}[\nabla P_{\mathrm{re}}; \nabla P_{\mathrm{im}}](d) = 2$, and *singular* (or tangential) otherwise.

Lemma 4.2 (Lean)**.** *For* $d \in \mathbb{R}^n$, $Df_A(d)$ *is not onto iff there is* $t \in \mathbb{R}$ *with* $(1-t^2) \operatorname{Re} \partial_i f_A(d) + 2t \operatorname{Im} \partial_i f_A(d) = 0$ *for all* i. *In particular a contact is singular iff it solves system (c).* $Df_A(d)$ *is onto iff some* 2×2 *minor* $\operatorname{Re} \partial_i f \operatorname{Im} \partial_j f - \operatorname{Re} \partial_j f \operatorname{Im} \partial_i f$ *is nonzero.*

Proof. All 2×2 minors vanish iff the vectors $(\operatorname{Re} \partial_i f, \operatorname{Im} \partial_i f)$ are all orthogonal to a common nonzero (p, q); then t is obtained from (p, q) as in Corollary 3.5. Conversely a common annihilator $(1-t^2, 2t) \neq 0$ forces every minor to vanish. □

Theorem 4.3 (Regular contacts persist; Lean)**.** *If d^* is a regular contact of A_0, then for every $\varepsilon > 0$ every matrix A in a neighbourhood of A_0 has a contact within distance ε of d^*. In particular no matrix near A_0 is D-stable.*

Proof. The map $F(A, d) = (A, f_A(d))$ is polynomial, and its derivative at (A_0, d^*) is $(\delta A, \delta d) \mapsto (\delta A, \partial_A f\, \delta A + \partial_d f\, \delta d)$, which is onto because $\partial_d f = D f_{A_0}(d^*)$ is onto. By the open mapping form of the implicit function theorem, F maps every neighbourhood of (A_0, d^*) onto a neighbourhood of $(A_0, 0)$. Taking the neighbourhood $\{(A, d) : |d - d^*| < \varepsilon,\ d > 0\}$, every A near A_0 has a contact d with $|d - d^*| < \varepsilon$, i.e. an imaginary eigenvalue of $A \operatorname{diag} d$. □

Corollary 4.4 (Lean)**.** *(a) Every contact of a matrix in the closure of the D-stable set is singular. (b) The Hurwitz matrices with a regular contact form an open set of matrices that are not D-stable.*

Theorem 4.5 (Boundary dichotomy; Lean)**.** *Let A_0 be Hurwitz and in the closure of the D-stable set. Then exactly one of the following holds.*

(i) A_0 is D-stable and $C_{A_0} = \emptyset$; moreover, for every $\kappa \geq 1$ all matrices near A_0 are κ-stable in the sense of Definition 6.1.

(ii) A_0 is not D-stable, $C_{A_0} \neq \emptyset$, and every contact of A_0 solves the tangency system (c).

Proof. If A_0 is D-stable, Lemma 2.3 gives $C_{A_0} = \emptyset$ and Theorem 6.4 gives the second statement. Otherwise A_0 has a contact; by Corollary 4.4(a) every contact is singular, and Lemma 4.2 applies. □

Case (i) holds for every D-stable matrix. When A_0 also lies on the frontier of the D-stable set we call it an *escape point*: it is a D-stable matrix outside the interior described by Hartfiel and Abed [13, 1], and the contacts of nearby non-D-stable matrices must escape towards the faces of the orthant (Theorem 6.4). Frontier points in case (ii) are *tangential points*.

Example 4.6 (The tangency branch cannot be dropped; exact)**.** Let

$$B_3 = -\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 2 & 0 & 1 \end{pmatrix} \operatorname{diag}(1, 2, 3) = \begin{pmatrix} -1 & -4 & 0 \\ 0 & -2 & -6 \\ -2 & 0 & -3 \end{pmatrix}.$$

Its characteristic polynomial is $\lambda^3 + 6\lambda^2 + 11\lambda + 54$, so it is Hurwitz; $-B_3 \in P_0^+$, and in Remark 3.7 $m_{123} = 54 = \alpha^2$ with $\alpha = 3\sqrt{6}$, in the attained case. So B_3 is not D-stable and its only contact is the Cauchy–Schwarz point $d^* = 3^{-1/2}(1, \frac{1}{2}, \frac{1}{3})$, where $\nabla P_{\mathrm{re}} = (2, 4, 6)$ and $\nabla P_{\mathrm{im}} = 3^{-1/2}(2, 4, 6)$ (rank one). For $a = b = \mathbf{1}$, $\nabla \psi_{a,b}(d^*) = (1 - \sqrt{3}, 1 - 2\sqrt{3}, 1 - 3\sqrt{3})$ is not a multiple of $(1, 2, 3)$, so system (b) has no solution while system (c) does. For $n = 5$, $A = \operatorname{diag}(B_3, -1, -1)$ has the contact set $\{d^*\} \times \mathbb{R}^2_{>0}$, on which $\partial_4 f_A = \partial_5 f_A = 0$; system (b) then forces $d_4 = d_5 = 1$ and the same contradiction in the first three coordinates. Replacing the entry -4 of B_3 by $-4+2\eta$ with small $\eta > 0$ lowers m_{123} to $54 - 24\eta$ and leaves α unchanged, so B_3 is a limit of D-stable matrices: a tangential point.

Example 4.7 (An isolated tangential contact at $n = 5$; exact and numerical)**.** Let $H(v) = I - 2vv^{\mathsf{T}}/(v^{\mathsf{T}}v)$, $Q = H(1, 2, 3, 4, 5)\, H(2, -1, 1, 3, -2)$ (rational orthogonal),

$$A_\varepsilon = Q\Big(\begin{pmatrix} 0 & -1/10 \\ 1/10 & 0 \end{pmatrix} \oplus \begin{pmatrix} -1 & 2+\varepsilon & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}\Big) Q^{\mathsf{T}}, \qquad \varepsilon = \tfrac{1}{1000},$$

and $D_0 = \operatorname{diag}(3/2, 4/5, 6/5, 7/10, 1)$. The matrix $B = D_0 A_\varepsilon$ is Hurwitz, and $d^* = 10 D_0^{-1}\mathbf{1} = (20/3, 25/2, 25/3, 100/7, 10)$ satisfies $f_B(d^*) = 0$ (since $B \operatorname{diag} d^*$ is similar to $10A_\varepsilon$) and

$\operatorname{rank}[\nabla P_{\mathrm{re}}; \nabla P_{\mathrm{im}}](d^*) = 1$ exactly. The Jacobian of system (c) has full column rank 6 at (d^*, t^*) (to 30 digits), so d^* is an isolated tangential contact; the parameter homotopy of Section 5 finds no real positive solution of (b) [numerical].

The set Σ_A of singular contacts need not be finite: for $A = \operatorname{diag}(B, C)$ with B (3×3) and C (2×2) Hurwitz and not D-stable, $f_A = f_B f_C$, so every point of $C_B \times C_C$ is singular and Σ_A contains a curve. The branch is nevertheless decided by a finite recursion.

Lemma 4.8 (Termination of the equality branch; conventional)**.** *Let $V \subseteq \mathbb{C}^n$ be an algebraic set defined over $\mathbb{R}$ and $\psi = \psi_{a,b}$. Put $Z_0 = V$ and $Z_{\ell+1} = \operatorname{Sing}(Z_\ell)$, the complex singular locus. Let Crit_ℓ be the set of points $d \in \mathbb{R}^n_{>0} \cap Z_\ell \setminus \operatorname{Sing}(Z_\ell)$ at which $\nabla\psi(d)$ lies in the span of the gradients of generators of the radical ideal of Z_ℓ (a space whose dimension is the local codimension of Z_ℓ at d). Then $V \cap \mathbb{R}^n_{>0} \neq \emptyset$ iff $\operatorname{Crit}_\ell \neq \emptyset$ for some $\ell \leq \dim V$. For weights (a, b) outside a proper algebraic subset every Crit_ℓ is finite.*

Proof. Each Crit_ℓ lies in $V \cap \mathbb{R}^n_{>0}$. Conversely let ℓ be the largest level with $Z_\ell \cap \mathbb{R}^n_{>0} \neq \emptyset$ and d^* a minimizer of ψ on this set (it exists as in Theorem 3.2). By maximality $d^* \notin \operatorname{Sing}(Z_\ell)$, so d^* is a smooth point of a unique irreducible component Y. Y is invariant under complex conjugation: otherwise $d^* = \overline{d^*}$ would lie on Y and on $\overline{Y} \neq Y$, hence in the singular locus. Near d^* the real points of Z_ℓ therefore form a real-analytic manifold cut out by real generators of the ideal of Y whose Jacobian has maximal rank, and the multiplier rule gives $d^* \in \operatorname{Crit}_\ell$. Since $\dim Z_{\ell+1} < \dim Z_\ell$ the recursion stops. The weights enter $\nabla\psi$ affinely and freely ($\partial_i \psi = a_i - b_i/d_i$), so the incidence of critical points is an affine bundle over the smooth locus, and generic smoothness gives finiteness for generic weights, as in [3, 30, 4]. Each Crit_ℓ is computed as a zero-dimensional system after saturation by the ideal of $\operatorname{Sing}(Z_\ell)$. □

Applied to $V = V_A \cap \{\text{all } 2 \times 2 \text{ minors of } [\nabla P_{\mathrm{re}}; \nabla P_{\mathrm{im}}] = 0\}$, the complex algebraic set whose positive real points form Σ_A, Lemma 4.8 decides branch (c) exactly; explicit degree bounds follow from Heintz's Bézout inequality [14]. For Example 4.7 the recursion stops at level 0 with $\operatorname{Crit}_0 = \{d^*\}$, and for the block example it stops at level 0 on the curve $C_B \times C_C$ [numerical].

5 Size and finiteness of the Lagrange system

Proposition 5.1 (Size law; exact)**.** *Group the unknowns of system (b) in the blocks $\{d_1\}, \ldots, \{d_n\}$ and $\{\lambda, \mu\}$. Its multihomogeneous Bézout number is $B(n) = n!\binom{n}{2}$: $2, 18, 144, 1200, 10800$ for $n = 2, \ldots, 6$.*

Proof. $P_{\mathrm{re}}, P_{\mathrm{im}}$ have multidegree $(1, \ldots, 1; 0)$, and each Lagrange equation has multidegree $(1, \ldots, 1; 1)$ because $d_i \partial_i P_{\mathrm{re}}$ is again multiaffine. The Bézout number is the coefficient of $z_1 \cdots z_n w^2$ in $(z_1 + \cdots + z_n)^2 (z_1 + \cdots + z_n + w)^n$: choose the two Lagrange factors that contribute w in $\binom{n}{2}$ ways, then assign the n distinct z_i to the remaining n factors in $n!$ ways. □

Table 2: Bézout numbers $B(n)$ (exact) and generic numbers $N(n)$ of complex solutions of system (b) (monodromy with random complex parameters, cross-checked by a linear-product homotopy for $n \leq 6$; numerical).

n	2	3	4	5	6	7
$B(n) = n!\binom{n}{2}$	2	18	144	1200	10800	105840
$N(n)$ (generic)	2	18	132	980	7830	68502

For $n = 2, \dots, 7$ the counts in Table 2 equal $n! \sum_{c=0}^{n-2} (n-1-c)/c!$ [numerical]. For every n this expression equals $(-1)^n \chi(V_n \setminus H)$, the signed Euler characteristic of the complex contact variety V_n with independent generic coefficients minus a generic hyperplane H [sketch]: V_n is then a nondegenerate complete intersection in $(\mathbb{C}^*)^n$ whose two equations have the even and the odd demicube of $[0,1]^n$ as Newton polytopes, Khovanskii's formula [21] writes $(-1)^n \chi(V_n \setminus H)$ as the sum of the mixed volumes $\mathrm{MV}(\Delta_{\mathrm{e}}^a, \Delta_{\mathrm{o}}^b, \Sigma^c)$ over $a, b \geq 1$, $a + b + c = n$, with Σ the standard simplex, and each of these equals $n!/c!$. Counting critical points by an Euler characteristic is in the spirit of [15]; that $N(n)$ equals it is proved for $n \leq 3$ and observed for $4 \leq n \leq 7$. If it holds in general, $N(n)/B(n) \sim 2e/n$ (not used in the proofs).

Proposition 5.2 (Generic finiteness; sketch)**.** *Let A be real and suppose P_{re} and P_{im} have no common nonconstant factor. For (a, b) outside a proper algebraic subset of $\mathbb{R}^{2n}$, every solution of (b) in $\mathbb{C}^{n+2}$ with d in the smooth part of V_A is nonsingular, so these solutions are finitely many (at most $B(n)$). If $g = \gcd(P_{\mathrm{re}}, P_{\mathrm{im}})$ is nonconstant, treat the hypersurface $\{g = 0\}$ by the one-multiplier system $\nabla\psi = \lambda \nabla g$, $g = 0$, and the rest by (b) for $P_{\mathrm{re}}/g, P_{\mathrm{im}}/g$.*

Proof sketch. In the i-th Lagrange equation b_i appears only in the term $-b_i$, and a_i only in $a_i d_i$. So the incidence variety $\{(d, \lambda, \mu, a, b)\}$ over the smooth part of V_A is an affine bundle, hence smooth of dimension $2n$. Its projection to (a, b) is generically finite and unramified by generic smoothness in characteristic zero. When $P_{\mathrm{re}}, P_{\mathrm{im}}$ are coprime, a dimension count excludes solutions over the singular locus for generic weights. □

Decision procedure. (1) Test the Hurwitz property exactly. (2) Split off $\gcd(P_{\mathrm{re}}, P_{\mathrm{im}})$. (3) Choose rational weights, compute a Gröbner basis of (b) saturated by $d_1 \cdots d_n$, and isolate its real roots; if it is not zero-dimensional, change the weights (a finite grid larger than the degree of the bad set contains a good choice). (4) Decide (c) by Lemma 4.8. This is a finite exact procedure whose correctness rests on Theorem 3.6 (Lean) and on Proposition 5.2 and Lemma 4.8. The size $B(n) = n!\binom{n}{2}$ grows super-exponentially with n.

What was run. The exact procedure has not been executed at $n = 5$: no Gröbner or real-root engine of the required scale was available. At $n = 3$ it was carried out exactly, with $a = b = \mathbf{1}$, for the matrix obtained from B_3 of Example 4.6 by replacing the entry -4 with $-21/5$ ($\eta = -1/10$). It is Hurwitz with $-A \in P_0^+$ and violates Cain's inequality ($m_{123} = 282/5 > 54 = \alpha^2$). After eliminating (λ, μ), resultants give a univariate eliminant of degree 24 in d_1 (square-free part of degree 20; the resultants add factors that do not correspond to solutions, so the degree exceeds $B(3) = 18$). Its real roots isolate exactly two real positive solutions of (b), and every equation vanishes at both to 40 digits. The implementation used in Sections 7–8 solves (b) by parameter homotopy from a start system of 980 solutions, uses branch (c) only through exact rational contacts found by a local search, and certifies every verdict independently: non-D-stability by an exact Routh count or an exact contact, D-stability by an exact diagonal Lyapunov certificate, an exact block-triangular certificate, or a rigorous exclusion (Section 8).

6 Time-scale robustness

Definition 6.1. For $\kappa \geq 1$ let $K_\kappa = \{d \in \mathbb{R}^n_{>0} : d_i \leq \kappa d_j \ \forall i, j\}$. A is *κ-stable* if $A \operatorname{diag} d$ is Hurwitz for every $d \in K_\kappa$. The *minimal destabilizing spread* is $\kappa^*(A) = \inf\{\kappa \geq 1 : A \text{ is not } \kappa\text{-stable}\}$, with $\kappa^* = \infty$ iff A is D-stable.

For $d \in \mathbb{R}^n_{>0}$ let $\mathrm{spread}(d) = \max_i d_i / \min_i d_i$; then $d \in K_\kappa$ iff $\mathrm{spread}(d) \leq \kappa$. In the settings of Table 1, κ bounds the ratio of abundances, hold-ups or loop gains, and κ^* is the smallest heterogeneity that destabilizes.

Theorem 6.2 (Bounded time-scale stability; Lean)**.** *Let $\kappa \geq 1$.*

(a) A is κ-stable iff A is Hurwitz and $C_A \cap K_\kappa = \emptyset$.

(b) Let $G(e,\omega) = \det(i\omega I - A\,\mathrm{diag}\,e)$ and let φ be any C^1 function on $\mathbb{R}^{n+1}$. Then A is κ-stable iff A is Hurwitz and there is no $(e,\omega) \in [1,\kappa]^n \times (0,\infty)$ with $G(e,\omega) = 0$ at which, for some $(\Lambda_1,\Lambda_2,\Lambda_0) \neq 0$, the functional $\Lambda_1 D\,\mathrm{Re}\,G + \Lambda_2 D\,\mathrm{Im}\,G + \Lambda_0 D\varphi$ vanishes on all directions v with $v_i = 0$ whenever $e_i \in \{1,\kappa\}$.

(c) A is D-stable iff it is κ-stable for every $\kappa \geq 1$, and κ'-stability implies κ-stability for $\kappa \leq \kappa'$.

Proof. (a) As in Lemma 2.3, with the connected set K_κ in place of $\mathbb{R}^n_{>0}$; rescaling a rate vector by $1/|\omega|$ keeps it in K_κ. (b) Every $d \in K_\kappa$ is e/ω with $e = d/\min_j d_j \in [1,\kappa]^n$ and $\omega = 1/\min_j d_j$, and $f_A(e/\omega) = \omega^{-n} G(e,\omega)$. If A is Hurwitz, the zero set of G in $[1,\kappa]^n \times [0,W]$ with $W = 1 + \kappa \sum_{ij} |a_{ij}|$ is compact and avoids $\omega = 0$ (there $G = \det(-A\,\mathrm{diag}\,e) \neq 0$) and $\omega = W$ (a row-sum bound on eigenvalues). A minimizer of φ on it lies in the relative interior of a unique face of the box; there it is a local minimizer with the active coordinates fixed, and the multiplier rule with the extra constraints $e_i = \text{const}$ gives the stated functional. (c) Every $d > 0$ lies in some K_κ. □

The faces of the box $[1,\kappa]^n$ are defined by independent coordinate constraints. The cone K_κ itself has linearly dependent active constraints when several rates tie at the maximum and at the minimum, which is why the box coordinates are used.

Proposition 6.3 (κ^* is attained; Lean. κ^* is algebraic; conventional)**.** *Let A be Hurwitz and not D-stable. (a) The infimum defining $\kappa^*(A)$ is a minimum: A is not κ^*-stable, it is κ-stable for every $1 \leq \kappa < \kappa^*$, and some contact lies in K_{κ^*}. Hence $\kappa^*(A) = \min\{\mathrm{spread}(d) : d \in C_A\}$. (b) $\kappa^*(A)$ is algebraic over the field generated by the entries of A.*

Proof. (a) Fix a failing κ_0. The triples (κ,e,ω) with $1 \leq \kappa \leq \kappa_0$, $e \in [1,\kappa]^n$, $\omega \in [0,W]$ and $G(e,\omega) = 0$ form a compact set; by Theorem 6.2(a) and the reduction to the box, its projection to κ is exactly the set of failing $\kappa \leq \kappa_0$, so it has a least element, and $\omega = 0$ is excluded because A is nonsingular. A contact in K_{κ^*} has spread at most κ^*, and a contact of smaller spread would lie in some K_κ with $\kappa < \kappa^*$. (b) By (a), κ^* is the least element of the set of $\kappa \geq 1$ for which the semialgebraic system $G(e,\omega) = 0$, $1 \leq e_i \leq \kappa$, $\omega > 0$ has a solution; this set is semialgebraic over the entries of A, so by Tarski–Seidenberg its least element is algebraic over them. The Fritz–John conditions on the faces of the box, with κ as an unknown, give a practical way to compute it. □

Because Definition 6.1 counts eigenvalues with $\mathrm{Re} \geq 0$, a contact of spread s gives $\kappa^* \leq s$ even when no scaling produces an eigenvalue with positive real part. For B_3 of Example 4.6 the unique contact gives $\kappa^*(B_3) = \mathrm{spread}(d^*) = 3$ exactly, while no scaling has an eigenvalue with positive real part, because B_3 is a limit of D-stable matrices.

Certified computation of κ^*. Upper bounds are exact: an exact Routh count > 0 at a rational rate vector of spread $\leq \kappa_{\mathrm{hi}}$, found by maximizing the spectral abscissa of $A\,\mathrm{diag}\,d$ over $[1,\kappa]^n$ and bisecting in κ, or an exact contact. Lower bounds are rigorous (Appendix A).

Theorem 6.4 (Separation protection; Lean)**.** *Let A_0 be Hurwitz with $C_{A_0} = \emptyset$ (for instance, A_0 D-stable). Then for every $\kappa \geq 1$ all matrices in a neighbourhood of A_0 are κ-stable. Equivalently: if $A_k \to A_0$ with A_0 Hurwitz and every A_k is destabilized within a fixed ratio bound κ, then A_0 has a contact.*

Proof. $G(A; e, \omega) = \det(i\omega I - A \operatorname{diag} e)$ is continuous in (A, e, ω) and does not vanish on $\{A_0\} \times [1, \kappa]^n \times [0, W]$: at $\omega = 0$ because A_0 is nonsingular, and for $\omega > 0$ because A_0 has no contact. By compactness (the tube lemma) $G(A; \cdot) \neq 0$ on the box for all A near A_0. The row-sum bound excludes $\omega > W$ uniformly near A_0, the Hurwitz matrices are open, and Theorem 6.2(a) concludes. □

Hence $\kappa^*(A) \to \infty$ as A approaches an escape point: such boundaries are *separation-protected*. At a tangential point A_0 the matrix itself has a contact d^*, so $\kappa^*(A_0) \leq \operatorname{spread}(d^*)$; under the nondegeneracy conditions of Proposition 6.5, nearby non-D-stable matrices are destabilized by bounded heterogeneity.

Proposition 6.5 (Tangential boundaries are heterogeneity-fragile; conventional)**.** *Let $A(\theta)$ be a C^2 family with a tangential point $A(\theta_0)$ and contact d^*. Assume (T1)* $\operatorname{rank} Df_{A(\theta_0)}(d^*) = 1$*; (T2) the Lyapunov–Schmidt reduced function $g(\theta, u)$, with $u \in \ker Df_{A(\theta_0)}(d^*) \cong \mathbb{R}^{n-1}$, has a definite Hessian in u at $(\theta_0, 0)$; (T3) $\partial_\theta g(\theta_0, 0) \neq 0$. Then on one side of θ_0 the matrices $A(\theta)$ have regular contacts $d(\theta)$ with $|d(\theta) - d^*| = O(|\theta - \theta_0|^{1/2})$, and $\kappa^*(A(\theta)) \leq \operatorname{spread}(d(\theta)) \to \operatorname{spread}(d^*)$.*

Proof. By (T1) the component of $f = 0$ along the range of Df can be solved, by the implicit function theorem, for one coordinate transversal to the kernel; the remaining scalar equation is $g(\theta, u) = 0$ with $g(\theta_0, 0) = 0$ and $\nabla_u g(\theta_0, 0) = 0$. By (T2) and (T3), $g(\theta, u) = \alpha(\theta - \theta_0) + \frac{1}{2} u^{\mathsf{T}} H u + o(|\theta - \theta_0| + |u|^2)$ with $\alpha \neq 0$ and H definite, so on the side where $\alpha(\theta - \theta_0)$ has the sign opposite to H the zero set in u is a small ellipsoid of radius $O(|\theta - \theta_0|^{1/2})$ around 0, and on the other side it is empty near 0. On the ellipsoid $\nabla_u g \neq 0$, so these contacts are regular. □

Remark 6.6 (Divergence at an escape point with one fast index; sketch)**.** Let $A(\theta)$ have an escape point at θ_0, let f be an index with $a_{ff}(\theta_0) < 0$, write $\bar{f}$ for the other indices, and let $S(\theta) = A_{\bar{f}\bar{f}} - A_{\bar{f}f} a_{ff}^{-1} A_{f\bar{f}}$ be the Schur complement of the fast index (the slow system when $d_f \to \infty$). Suppose $S(\theta_0)$ has a nondegenerate tangential contact e^* at frequency 1, with right and left eigenvectors v, w of $S(\theta_0) \operatorname{diag} e^*$ for i. With $\alpha = \partial_\theta \operatorname{Re} \lambda(S(\theta) \operatorname{diag} e^*)|_{\theta_0}$ (the eigenvalue continuing i) and $\beta = \operatorname{Re}\big(i\, w^* A_{\bar{f}f} A_{f\bar{f}} \operatorname{diag}(e^*) v / (a_{ff}^2 w^* v)\big)$, the slow eigenvalues of $A \operatorname{diag}(e, L)$ solve $\det(S_\varepsilon(\lambda) \operatorname{diag} e - \lambda) = 0$ with $\varepsilon = 1/L$ and $S_\varepsilon(\lambda) = A_{\bar{f}\bar{f}} - A_{\bar{f}f}(a_{ff} - \varepsilon\lambda)^{-1} A_{f\bar{f}} = S - \varepsilon\lambda A_{\bar{f}f} A_{f\bar{f}} / a_{ff}^2 + O(\varepsilon^2)$, so near e^* $\operatorname{Re}\lambda \approx \alpha\delta - \beta\varepsilon$ at $\theta = \theta_0 + \delta$. If $\alpha, \beta > 0$ and no other face of the orthant destabilizes, this suggests $\kappa^*(A(\theta_0 + \delta)) \approx (\beta/\alpha)\, \delta^{-1} / \min_i e_i^*$, i.e. divergence exponent one. We have not proved the matching lower bound in general; Example 6.7 and Table 6 support the prediction.

One-parameter families. For a polynomial family $A(\theta)$ the set of θ with $A(\theta)$ D-stable is semialgebraic, hence a finite union of points and intervals. By Theorem 4.5 each endpoint at which $A(\theta)$ is Hurwitz is either tangential (a root of the eliminant of system (c) with θ free) or an escape point. For escape points we use *face systems*: for disjoint index sets F (fast) and Z (frozen), the principal submatrix, on the indices outside $F \cup Z$, of the Schur complement of $A[F]$. These are the limits of the slow part of $A \operatorname{diag} d$ when the rates in F tend to infinity and those in Z to zero. Generically an escape point is an endpoint of D-stability of a face system of lower order, which gives a finite candidate set; we use this as a procedure, not a theorem (face systems that are identically marginal on an interval need higher-order terms).

Example 6.7 (Automatic recovery of a known window; exact)**.** The 4×4 family $B(h)$ of [33] (rows $(-1, -\frac{7}{10}, -\frac{7}{10}, -\frac{14}{5})$, $(4, -1, -\frac{7}{10}, -\frac{14}{5})$, $(4, 4, -1, -\frac{14}{5})$, $(-1, -1, -1, -h)$) is known, for $h \geq 4$, to be D-stable exactly when $K(h) = 3890h^2 - 38717h + 93492 \geq 0$. Generating candidates on $[2, 9]$ from the Hurwitz conditions, the principal 3×3 submatrices and the four

single-index Schur complements (each tested by Cain's criterion) yields, among others, exactly $h_\pm = (38717 \pm 329\sqrt{409})/7780$, as roots of Cain's equality for the Schur complement with the fourth index fast; $K(h)$ divides the rationalized Cain resolvent of that Schur complement. Deciding one rational sample point between consecutive candidates, with every verdict certified (exact Routh test, exact Routh count at a rational scaling, or rigorous exclusion), reproduces the window and places the only other change of verdict on $[2, 9]$ at the Hurwitz boundary $h = 18613/4660 \approx 3.9942$, where $\det(-B(h))$ vanishes. At both window edges κ^* diverges like δ^{-1} (Table 6), and the constant of Remark 6.6 agrees: at $h_\mp$ the Schur complement's tangential contact is the Cauchy–Schwarz point of Remark 3.7, and $(\beta/\alpha)/\min_i e_i^* = 3.509$ and 8.809, against the measured $\kappa^*\delta = 3.511$ and 8.815 at $\delta = 10^{-4}$ [numerical].

7 Worked examples

7.1 A five-species Lotka–Volterra community

Consider the Lotka–Volterra community

$$\dot{x}_i = x_i(r_i + s_i((J - \varepsilon I)x)_i), \qquad i = 1, \dots, 5,$$

with the cyclic interaction matrix $J = \mathrm{circ}(-\frac{6}{5}, 2, -\frac{2}{5}, -\frac{2}{5}, 0)$ (self-limitation, one strong positive and two weak negative neighbour effects), self-regulation $\varepsilon \geq 0$ and species sensitivities $s = (1, 2, 3, 4, 5)$. The interaction matrix is $A(\varepsilon) = S(J - \varepsilon I)$ with $S = \mathrm{diag}\, s$, and every feasible coexistence equilibrium is stable iff $A(\varepsilon)$ is D-stable (Table 1). The model is stylized; all numbers below are computed from these inputs.

Proposition 7.1 (exact)**.** *Let* $\varepsilon_H = (7\sqrt{5} - 15)/10 \approx 0.06525$.

(a) $A(\varepsilon)$ *is D-stable iff* $\varepsilon > \varepsilon_H$*, for every sensitivity vector* $s > 0$.

(b) For every $\varepsilon \geq 0$*, every* 4×4 *principal submatrix of* $A(\varepsilon)$ *is D-stable: every four-species subcommunity is stable at all abundances.*

(c) At $\varepsilon \in \{1/1000, 1/100, 3/100, 3/50, 13/200\} \subset (0, \varepsilon_H)$ *the equilibrium with equal abundances is stable while the equilibrium* $x^* = S^{-1}\mathbf{1}$ *is unstable, and* $A(\varepsilon_H)$ *is Hurwitz; numerically, stability at equal abundances holds on all of* $(0, \varepsilon_H]$.

Proof. (a) J is circulant, hence normal, with eigenvalue real parts 0, ε_H (twice) and $-(15 + 7\sqrt{5})/10$ (twice). For $\varepsilon > \varepsilon_H$, $P = S^{-1}$ gives $PA + A^{\mathsf{T}}P = (J - \varepsilon I) + (J - \varepsilon I)^{\mathsf{T}} \prec 0$, so $A(\varepsilon)$ is diagonally stable, hence D-stable. For $\varepsilon \leq \varepsilon_H$ the abundances $x^* = S^{-1}\mathbf{1}$ give the Jacobian $J - \varepsilon I$, which has an eigenvalue with nonnegative real part. (b) Each 4×4 principal submatrix J_I of J has a rational diagonal Lyapunov certificate $P_I J_I + J_I^{\mathsf{T}} P_I \prec 0$ (exact; the five certificates are cyclic shifts of $\mathrm{diag}(\frac{1}{2}, \frac{11}{10}, 1, \frac{19}{10})$). It persists when εI is subtracted, and D-stability is invariant under left multiplication by S_I. (c) Exact Routh counts; for ε_H, exact Hurwitz minors in $\mathbb{Q}(\sqrt{5})$. □

Normal interaction matrices are thus D-stable as soon as they are stable at equal abundances; heterogeneous sensitivities break normality. On $(0, \varepsilon_H]$ the community is stable at equal abundances but not robust. Table 3 and Figure 2 give $\kappa^*(\varepsilon)$, the least abundance ratio $\max_i x_i^* / \min_i x_i^*$ at which some feasible equilibrium is unstable. It increases from about 1.47 and approaches $5 = \mathrm{spread}(S^{-1}\mathbf{1})$ as $\varepsilon \uparrow \varepsilon_H$. The matrix $A(\varepsilon_H)$ is Hurwitz, a limit of D-stable matrices and has the contact $S^{-1}\mathbf{1}/\omega$ for a suitable $\omega > 0$, so it is a tangential point (Theorem 4.5). At $\varepsilon = \varepsilon_H/2$ the destabilizing abundances are, relative to the rarest species, $(1.96, 1.96, 1.11, 1.00, 1.00)$ for species $1, \dots, 5$.

Table 3: The community of Section 7.1: minimal destabilizing abundance ratio. Upper bound: exact Routh count at a rational abundance vector of that spread. Lower bound: rigorous exclusion of contacts from $K_{0.97\kappa^*}$ (Appendix A). The last row is $\varepsilon = 26099/400000 \approx \varepsilon_H - 8.4 \cdot 10^{-8}$.

$\varepsilon/\varepsilon_H$	ε	κ^* (num.)	certified upper	certified lower
0.02	0.0013050	1.4669	1.467	1.4229
0.1	0.0065248	1.5266	1.527	1.4808
0.25	0.0163119	1.6576	1.658	1.6079
0.5	0.0326238	1.9631	1.963	1.9042
0.75	0.0489357	2.5496	2.55	2.4731
0.9	0.0587228	3.2241	3.225	3.1274
0.97	0.0632902	3.9314	3.932	3.8134
0.999999	0.0652475	4.9921	4.993	4.8424

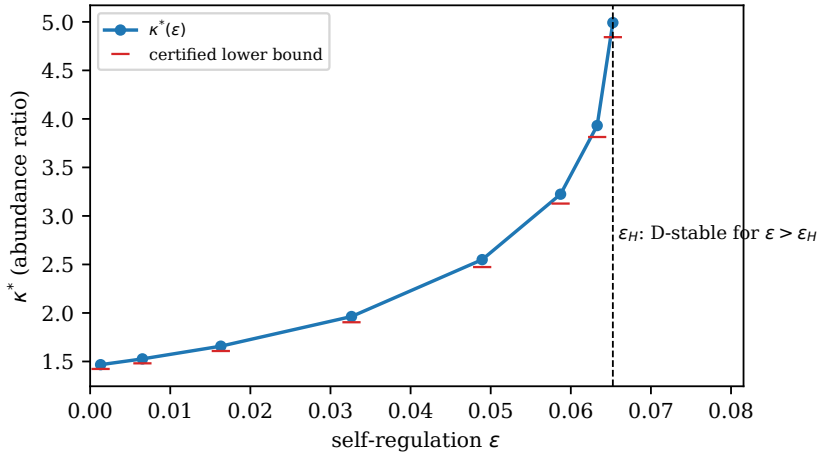


Figure 2: Section 7.1: $\kappa^*(\varepsilon)$ on $(0, \varepsilon_H]$ with the certified lower bounds. D-stability holds exactly for $\varepsilon > \varepsilon_H$; stability at equal abundances holds on the whole interval shown.

7.2 A published refinery-column gain matrix

Consider a stable, finite-dimensional, strictly proper plant $G(s)$ with integral loops k_i/s, negative feedback, and the column sign normalization $G^+(0) = G(0)\,\mathrm{diag}(\mathrm{sign}\, g_{ii}(0))$. Campo and Morari [8] prove that the plant is decentralized integral controllable (DIC) if every principal submatrix of $-G^+(0)$ is D-stable in our convention (Thm. 8), and that DIC requires every principal submatrix of $-G^+(0)$ to be nonsingular and weakly D-stable, i.e. every scaling DM with $D > 0$ has its spectrum in the closed left half-plane (Thm. 7); D-stability of $G(0)$ alone is not equivalent to DIC [8, Ex. 3]. In the low-gain regime the slow closed-loop modes follow $\dot{x} = -G^+(0)Kx$ with K the diagonal gain matrix, so $\kappa^*(-G^+(0))$ is the least loop-gain ratio that destabilizes them.

We use the 4×4 steady-state gain matrix of an industrial naphtha column at the Al-Doura refinery, identified by pulse tests [25, eq. (27)]. Two caveats apply. Its input X_f (feed composition) is a disturbance rather than a manipulated variable, and the model has dead times, which lie outside the finite-dimensional setting of [8]. We therefore report total D-stability of $-G^+(0)$ for the published 4×4 matrix, and interpret it as DIC only for the 3×3 subsystem of manipulated inputs (R, Q_R, F) paired with (X_D, X_B, B), and only for the delay-free model with the same steady-state gain. As a check we also treat the sidestream column of Elaahi and Luyben [10], as reprinted in [32, Ex. 3], known not to be DIC for the

diagonal pairing; the eigenvalues of $G^+(0)$ computed from the reprinted entries agree with the values printed there. No verifiable 5×5 gain matrix of a real process was found.

Table 4: Total D-stability of $-G^+(0)$ (diagonal pairing). 1×1, 2×2 (the P_0^+ test [17]) and 3×3 (Cain [7]) principal submatrices are decided exactly. A diagonal Lyapunov certificate for the whole matrix implies D-stability of every principal submatrix, because it restricts to them; for the Al-Doura 4×4 matrix this implication and the certificate are checked in Lean.

Gain matrix	eigenvalues of $G^+(0)$	principal submatrices of $-G^+(0)$ not D-stable	verdict
Al-Doura naphtha column, 4×4 [25]	1.25, 1.61, $17.86 \pm 3.45i$	none	totally D-stable; diagonal Lyapunov certificate $P = \mathrm{diag}(2/5, 2/5, 81/10, 4/5)$ (Lean); $\kappa^* = \infty$
same, manipulated inputs (R, Q_R, F) only, 3×3	0.79, 1.65, 23.15	none	totally D-stable; diagonal Lyapunov certificate $P = \mathrm{diag}(7/10, 4/5, 9/5)$; $\kappa^* = \infty$
Elaahi–Luyben sidestream column, 4×4 [10, 32]	-9.69, 4.74, 6.05, 19.88	{1,3}; {1,4}; {2,4}; {3,4}; {1,2,3}; {1,2,4}; {1,3,4}; {2,3,4}; {1,2,3,4}	not totally D-stable; every listed submatrix is not even Hurwitz, so $\kappa^* = 1$ for it

7.3 Known families as boundary types

Table 5 places the benchmark families on the two sides of the dichotomy of Theorem 4.5. Figure 3, Table 6 and Figure 4 describe the certified boundary atlas of Section 8. There a bracket is called a *face birth* if its unstable endpoint has an exact face certificate (an exact Routh count > 0 at a scaling with one group of rates 10^k times the others), and an *interior birth* otherwise; these are heuristic proxies for escape and tangential boundary points, not certified classifications. Along interior-birth segments κ^* stays bounded ($p \approx 0$ in Table 6). Along face-birth segments it diverges: 20 of the 21 segments give p between 0.81 and 1.07, as Remark 6.6 predicts for one fast index (the same exponent is observed for larger fast sets), and one random face-birth segment grows faster, by a factor of about 16 per factor 4 in δ over most of its range ($p \approx 2$; its fitted value 3.22 is driven by the bound at the smallest distance). Faster growth is expected when the crossing of the slow system is degenerate ($\alpha = 0$ in Remark 6.6).

7.4 Runtime

Table 7 reports the wall-clock time of the complete 5×5 decision (compiled parameter homotopy for (b), exact face checks, and an independent certificate) on 120 certified boundary instances of the atlas, on a shared workstation. All 120 verdicts agree with the independent certification of the atlas; the 60 D-stable verdicts are certified by 39 exact diagonal Lyapunov certificates and 21 rigorous exclusions, the 60 others by exact Routh counts.

8 Validation of the implementation

Test matrices. The calibration suite consists of 27 matrices of order 5 with independently known verdicts: 3×3 matrices on or near the boundary of Cain's criterion, embedded block-diagonally, weakly coupled or bordered; the family $B(h)$ of Example 6.7 at $h = 4, 5, 6$, bor-

Table 5: Boundary types of benchmark families.

Family	boundary type	evidence
$B(h)$ window edges $h_\pm$ [33]	escape	$h_\pm$ recovered exactly as roots of Cain's equality for a Schur complement; $\kappa^* \sim \delta^{-1}$ (Table 6)
coNP gadgets (Pell cores with one attachment) [34]	escape (heuristic)	exact face certificates at 40 of 40 certified births; κ^* grows like δ^{-1} (Table 6)
B_3 (Example 4.6)	tangential	exact: unique contact, of rank one; $\kappa^* = 3$
$D_0 A_\varepsilon$, $\varepsilon = 1/1000$ (Example 4.7)	tangential contact	exact contact of rank one; isolated (Jacobian rank 6)
community $S(J - \varepsilon I)$ at ε_H (Section 7.1)	tangential	$\kappa^* \to 5$ as $\varepsilon \uparrow \varepsilon_H$
circulant $J - \varepsilon I$ (normal)	–	D-stable iff Hurwitz; $P = I$ certifies for $\varepsilon > \varepsilon_H$

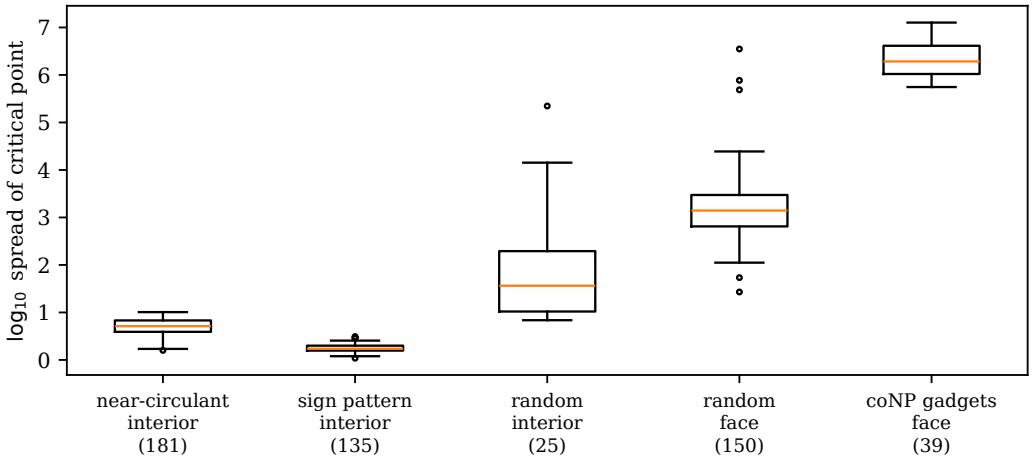


Figure 3: Certified boundary atlas: spread of the ψ-minimal critical point at the unstable endpoint of a certified bracket (an upper bound for κ^*), by matrix class and birth type, for the 530 of the 538 brackets of Section 8 whose critical point was recorded; counts in parentheses. The classes are defined in Section 8.

dered; vertex and off-vertex matrices of the coNP-hardness gadgets of [34]; perturbations of the circulant core J of Section 7.1, including Example 4.7; matrices with negative definite symmetric part; matrices with a non-D-stable 3×3 principal submatrix; and a family with symmetric principal minors. The boundary atlas uses segments $t \mapsto (1-t)A_s + tA_u$ from an exactly certified D-stable A_s to an exactly certified non-D-stable A_u in four classes: *random* ($A_s = -(MM^\mathsf{T} + I) + W - W^\mathsf{T}$ with small random integer matrices M, W, which is diagonally stable, and a random A_u); *near-circulant* ($D_0(J_q - \frac{1}{100}I)$ with a random positive diagonal D_0 and a circulant J_q moved from a D-stable to an unstable parameter); *sign pattern* (negative diagonal, $a_{i,i+1} > 0$, $a_{i+2,i} < 0$ cyclically, random magnitudes); and *coNP gadgets* (Pell cores with one attachment [34]). A *certified bracket* is a subsegment of relative width at most 10^{-3} whose endpoints are certified D-stable and non-D-stable independently of the criterion.

Calibration. All 27 calibration matrices are decided correctly, with 27 certified verdicts and 0 disagreements.

Table 6: Divergence of κ^* at distance δ from atlas boundary points, one row per birth type: exponent p in $\kappa^* \asymp \delta^{-p}$, fitted per segment to the certified upper bounds (exact Routh certificates) at the four smallest distances $\delta \in \{2^{-2}, 2^{-4}, \ldots, 2^{-20}\}$ where a certificate exists; values of κ^* above about 10^7 could not be certified in double precision and are not used.

Segments	count	median p	range of p
face births, coNP gadgets	10	0.99	[0.81, 1.03]
face births, random	11	1.00	[0.83, 3.22]
interior births, near-circulant	10	0.01	[0.00, 0.01]
interior births, sign pattern	10	0.02	[0.02, 0.02]
$B(h)$ at h_- (Example 6.7)	1	0.971	–
$B(h)$ at h_+ (Example 6.7)	1	0.995	–

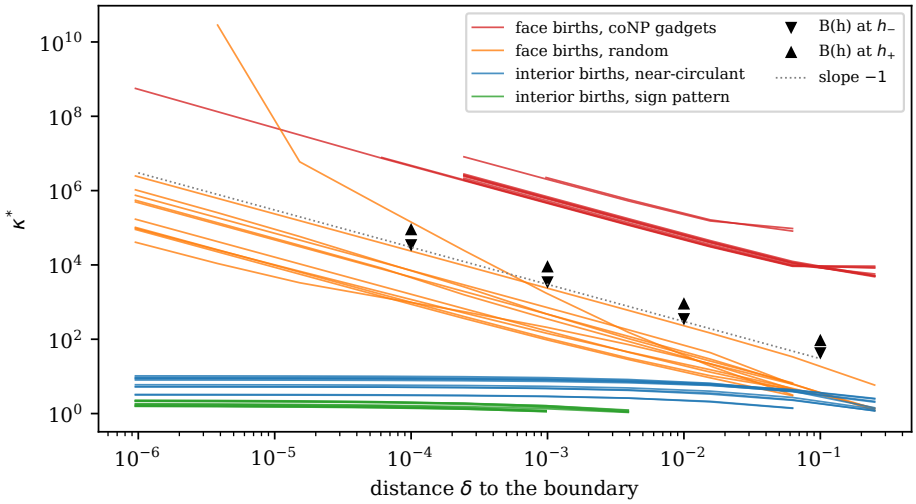


Figure 4: κ^* (certified upper bounds) versus the distance δ to the boundary on the atlas segments of Table 6: along face-birth segments κ^* diverges, in all but one case like δ^{-1}; along interior-birth segments it stays bounded. Black markers: the $B(h)$ window edges of Example 6.7.

Boundary atlas. In a first design (70 segments, 69 certified brackets), bisection used a multistart growth-rate sampler; the criterion found real positive critical points at 23 points that the sampler had labelled stable, and an exact Routh count confirmed strict D-instability at every one of them. In a second design (473 segments, 472 certified brackets, 3 of them certified after the run by a rigorous exclusion with a larger budget) bisection used the criterion's own verdict (compiled parameter homotopy), and every bracket endpoint was certified independently, in the direction the criterion predicted. Of the 541 certified brackets, the stable endpoints carry 363 exact diagonal Lyapunov certificates and 178 rigorous exclusions (Appendix A); the unstable endpoints carry 536 exact Routh counts, and 5 are not Hurwitz. With the 23 corrected points this gives 1105 certified instances concentrated at the boundary, with no unresolved disagreement.

One genuine disagreement. In one further bracket of the second design the implementation reported no critical point at an endpoint that is in fact strictly D-unstable (exact Routh certificate at $d = (1, 1, 1, 10^8, 6{\cdot}10^7)$): a Schur-complement face of a coNP-gadget matrix places the critical point at spread $\gtrsim 10^8$, beyond double-precision path tracking. The independent certification refused the bracket, so it is not among the certified brackets. The implementation

Table 7: Time per 5×5 decision (seconds).

Instances	count	median	95th percentile
near-circulant, stable endpoint	15	3.91	4.90
near-circulant, unstable endpoint	15	0.54	0.96
sign pattern, stable endpoint	15	3.60	4.72
sign pattern, unstable endpoint	15	0.84	1.00
random, stable endpoint	15	3.01	13.26
random, unstable endpoint	15	0.70	1.07
coNP gadgets, stable endpoint	15	2.76	3.18
coNP gadgets, unstable endpoint	15	0.63	2.21
all	120	2.26	4.85

now runs an exact face test before any stability certificate: for each index set F it evaluates an exact Routh count at $d_F = 10^k$ ($k \le 12$), which can only certify instability. The defect is in the floating-point implementation, not in Theorem 3.4, whose critical point exists by Theorem 3.2.

Invariance and $n = 3$. The Lagrange verdict is the same for three exhaustions (log barrier with unit and random weights, reciprocal) on 23/24 Hurwitz calibration matrices; the exception is a path-tracking miss of the reciprocal system at spread $5 \cdot 10^4$; direct constrained minimization of the reciprocal exhaustion on C_A recovers the missed critical point. The criterion reproduces Cain's exact 3×3 criterion on 60/60 random matrices embedded as $\mathrm{diag}(A_3, -1, -2)$.

Births. Of the 538 distinct brackets certified during the runs, 192 have an exact face certificate at the unstable endpoint (an exact Routh count > 0 at $d_F = 10^k$, $k \le 12$, for some index set F) and 346 do not; we call them face and interior births. Face births occur in the random (152 of 177) and coNP-gadget (40 of 40) classes; all near-circulant (181) and sign-pattern (140) births are interior births.

Census. On the 532 certified brackets of the validation runs (pilot runs excluded) the compiled homotopy found a real positive critical point at every one of the 527 Hurwitz unstable endpoints (the other unstable endpoints are not Hurwitz) and none at any of the 532 stable endpoints. On the 24 Hurwitz calibration matrices it found one for every non-D-stable matrix except Example 4.7, which branch (c) decides, and for no D-stable matrix.

9 Limitations and open problems

- *Decision procedure.* Lean verifies the criteria as equivalences. The finite procedure also uses Proposition 5.2 (sketch) and Lemma 4.8 (conventional) and has not been executed at $n = 5$; our verdicts come from floating-point homotopy together with independent exact or rigorous certificates, and branch (c) is used only through exact contacts. Double-precision path tracking loses critical points of spread $\gtrsim 10^8$, which occur near faces of the orthant; the implementation compensates with exact face checks. An exact run at $n = 5$ would need a zero-dimensional solver (Gröbner basis and real-root isolation) for (b), which has at most 1200 isolated solutions for generic weights, and the recursion of Lemma 4.8 for (c), whose radicals, singular loci and saturations are likely the costlier part; we have not attempted it.

- *Formalization.* Proposition 5.2, Proposition 5.1 as a bound on solutions, Lemma 4.8 and the decision procedure are not formalized: Mathlib lacks algebraic generic smoothness, multihomogeneous Bézout bounds, dimension theory of singular loci and real root isolation.
- *Divergence exponent.* Remark 6.6 is a sketch without a proved lower bound.
- *Finiteness of tangency sets.* We do not know whether the real tangency set of a Hurwitz matrix with coprime $P_{\mathrm{re}}, P_{\mathrm{im}}$ is finite; Lemma 4.8 makes this an efficiency question.
- *Complexity of bounded-ratio stability.* Is κ-stability coNP-hard in n for fixed κ?
- *Deficiency.* Does a 5×5 D-stable matrix exist all of whose proper principal submatrices of order at least 2 fail to be D-stable?
- *Nonlinear onset.* The results concern linear stability of equilibria.

10 Formal verification

The full development has root `Paper.lean`, under `proofs/DStability5x5/`. The manuscript's source verification used pinned Mathlib with Lean v4.30.0. The root and its import closure comprise 47 local modules: 14 in `proofs/DStability5x5/`, 23 in `proofs/DUnstableCores/`, one in `proofs/DStabilityCharacterization/` and 9 in three further directories. The basic notions (`DStable`, `HurwitzStable`, `HasEigenpair`, right scaling) come from `DScaling.lean` in `proofs/DUnstableCores/`. The continuation lemma is in `SpectralContinuation.lean`, under `proofs/DStabilityCharacterization/`. Several modules are unrelated to this paper and are compiled only because of imports. Each module was compiled with warnings treated as errors and without `sorry`; the exported declarations use only the axioms `propext`, `Classical.choice` and `Quot.sound`. D-stability is formalized literally: for every $d > 0$, every eigenpair of $A \operatorname{diag} d$ over $\mathbb{C}$ has negative real part. Statements marked conventional, sketch, exact or numerical are not formalized.

Formalization package. The accompanying repository contains the complete 39-module proof closure for the two selected theorems under `proofs/`, the two selected statements under `Registry/P087/`, and their correspondence to Theorems 3.4 and 3.6 under `entries/P087/`. The target toolchain and dependency revision are specified by `lean-toolchain` and `lake-manifest.json`. Numerical experiments and decision-tool execution are separate from the selected formal statements.

A Rigorous exclusion of contacts

For each of the 2^n charts $d_i = t_i$ or $d_i = 1/t_i$ with $t \in [0,1]^n$, multiplying f_A by the product of the t_i of the inverted coordinates gives a multiaffine polynomial $\tilde{f}$ in t whose values at the corners of $[0,1]^n$ are monomials of f_A (for instance i^n and $\det(-A)$); every positive d, including the limits $d_i \to 0$ and $d_i \to \infty$, is covered. By the multiaffine mapping theorem, $\tilde{f}(\mathrm{box})$ lies in the convex hull of the values of $\tilde{f}$ at the 2^n vertices of the box. A box is discarded if a unit vector w satisfies $\langle w, \tilde{f}(v)\rangle > \sqrt{2}\,\mathrm{err}(v)$ at every vertex v, where $\mathrm{err}(v) = 64u \sum_S |c_S| t^S$ bounds the floating-point error of evaluating $\tilde{f}(v)$ with unit roundoff u (the coefficients c_S are the principal minors, rounded once). Otherwise the box is bisected along its longest side. For a κ-stability certificate, boxes that contain no point of K_κ (some ratio d_i/d_j exceeds κ on the whole box) are pruned. If every box is discarded and A is Hurwitz (exact Routh test), A is κ-stable (or D-stable, without pruning) by Theorem 6.2(a). This procedure describes the numerical exclusion method used in the experiments.

Table 8: Lean correspondence (namespace `DStability5x5`).

Statement	Declaration	Module
Lemma 2.2	`contactDet_expansion`, `pminor_eq_minorR`	ContactExpansion, PrincipalMinor
Lemma 2.3	`dStable_iff_no_contact`	Cone
Theorem 3.2	`dStable_iff_morseFor`, `dStable_iff_morse_logBarrier`	ProperExhaustion
Theorem 3.4	`dStable_iff_explicit`	ExplicitCriterion
Corollary 3.5	`dStable_iff_lagrange_tangency`	FiveByFive
Theorem 3.6	`dStable_iff_five`	FiveByFive
Lemma 4.2	`not_surjective_iff_tangency`, `regular_iff_rank`	Boundary, Robustness
Theorem 4.3	`exists_contact_near_of_regular`, `not_dStable_near_of_regular`	Robustness
Corollary 4.4	`not_regular_of_mem_closure`, `isOpen_hurwitz_regularContact`, `hurwitz_regularContact_open_not_dStable`	Robustness
Theorem 4.5	`closure_dichotomy`	Boundary
Theorem 6.2	`kStable_iff_no_cone_contact`, `kStable_iff_no_boxFaceCritical`, `dStable_iff_forall_kStable`, `kStable_mono`	Cone
Proposition 6.3(a)	`exists_least_unstable_ratio`	KappaStar
Theorem 6.4	`eventually_kStable_of_no_contact`, `contact_of_tendsto_not_kStable`	Separation
Table 4	`dStable_of_diagLyapunov`, `alDoura_dStable`	Lyapunov, AlDoura

References

[1] Eyad H. Abed. Strong D-stability. *Systems & Control Letters*, 7(3):207–212, 1986.

[2] Kenneth J. Arrow and Maurice McManus. A note on dynamic stability. *Econometrica*, 26(3):448–454, 1958.

[3] Philippe Aubry, Fabrice Rouillier, and Mohab Safey El Din. Real solving for positive dimensional systems. *Journal of Symbolic Computation*, 34(6):543–560, 2002.

[4] Saugata Basu, Richard Pollack, and Marie-Françoise Roy. *Algorithms in Real Algebraic Geometry*, volume 10 of *Algorithms and Computation in Mathematics*. Springer, Berlin, Heidelberg, 2nd edition, 2006.

[5] Jérémy Berthomieu, Edern Gillot, and Mohab Safey El Din. Computing points in connected components defined by a real inequation: algorithms, complexity and implementations, Part I. arXiv:2605.18110 [cs.SC], 2026.

[6] Larisa A. Burlakova. Conditions of D-stability of the fifth-order matrices. In *Computer Algebra in Scientific Computing (CASC 2009)*, volume 5743 of *Lecture Notes in Computer Science*, pages 54–65, Berlin, Heidelberg, 2009. Springer.

[7] Bryan E. Cain. Real, 3×3, D-stable matrices. *Journal of Research of the National Bureau of Standards, Section B: Mathematical Sciences*, 80B(1):75–77, 1976.

[8] Peter J. Campo and Manfred Morari. Achievable closed-loop properties of systems under decentralized control: conditions involving the steady-state gain. *IEEE Transactions on Automatic Control*, 39(5):932–943, 1994.

[9] G. W. Cross. Three types of matrix stability. *Linear Algebra and its Applications*, 20(3):253–263, 1978.

[10] Abolghasem Elaahi and William L. Luyben. Control of an energy-conservative complex configuration of distillation columns for four-component separations. *Industrial & Engineering Chemistry Process Design and Development*, 24(2):368–376, 1985.

[11] Giorgio Giorgi and Cesare Zuccotti. An overview on D-stable matrices. DEM Working Paper Series 97 (02-15), Department of Economics and Management, University of Pavia, February 2015.

[12] D. Yu. Grigor'ev and N. N. Vorobjov, Jr. Solving systems of polynomial inequalities in subexponential time. *Journal of Symbolic Computation*, 5(1–2):37–64, 1988.

[13] D. J. Hartfiel. Concerning the interior of the D-stable matrices. *Linear Algebra and its Applications*, 30:201–207, 1980.

[14] Joos Heintz. Definability and fast quantifier elimination in algebraically closed fields. *Theoretical Computer Science*, 24(3):239–277, 1983.

[15] June Huh. The maximum likelihood degree of a very affine variety. *Compositio Mathematica*, 149(8):1245–1266, 2013.

[16] Serkan T. Impram, Russell Johnson, and Raffaella Pavani. The D-stability problem for 4×4 real matrices. *Archivum Mathematicum (Brno)*, 41(4):439–450, 2005.

[17] Charles R. Johnson. Second, third, and fourth order D-stability. *Journal of Research of the National Bureau of Standards, Section B: Mathematical Sciences*, 78B(1):11–13, 1974.

[18] Charles R. Johnson. A characterization of the nonlinearity of D-stability. *Journal of Mathematical Economics*, 2(1):87–91, 1975.

[19] Russell Johnson and Alberto Tesi. On the D-stability problem for real matrices. *Bollettino dell'Unione Matematica Italiana, Serie 8*, 2-B(2):299–314, 1999.

[20] G. V. Kanovei and D. O. Logofet. D-stability of 4-by-4 matrices. *Computational Mathematics and Mathematical Physics*, 38(9):1369–1374, 1998.

[21] A. G. Khovanskii. Newton polyhedra and the genus of complete intersections. *Functional Analysis and Its Applications*, 12(1):38–46, 1978.

[22] Olga Y. Kushel. Unifying matrix stability concepts with a view to applications. *SIAM Review*, 61(4):643–729, 2019.

[23] Olga Y. Kushel. Recursive determinantal framework for testing D-stability. I. arXiv:2604.16526 [math.SP], 2026.

[24] Olga Y. Kushel and Raffaella Pavani. The problem of generalized D-stability in unbounded LMI regions and its computational aspects. *Journal of Dynamics and Differential Equations*, 34(1):651–669, 2022.

[25] Layth S. Mahmood and Khalid M. Mousa Al-zobai. Modelling of a 4×4 multivariable distillation column using pulse function-based system identification. *Al-Khwarizmi Engineering Journal*, 22(3):111–123, 2026.

[26] Raffaella Pavani. A new efficient approach to the characterization of D-stable matrices. *Mathematical Methods in the Applied Sciences*, 41(12):4407–4416, 2018.

[27] James Renegar. On the computational complexity and geometry of the first-order theory of the reals. Part I: Introduction. Preliminaries. The geometry of semi-algebraic sets. The decision problem for the existential theory of the reals. *Journal of Symbolic Computation*, 13(3):255–299, 1992.

[28] James Renegar. On the computational complexity and geometry of the first-order theory of the reals. Part II: The general decision problem. Preliminaries for quantifier elimination. *Journal of Symbolic Computation*, 13(3):301–327, 1992.

[29] James Renegar. On the computational complexity and geometry of the first-order theory of the reals. Part III: Quantifier elimination. *Journal of Symbolic Computation*, 13(3):329–352, 1992.

[30] Mohab Safey El Din and Éric Schost. Polar varieties and computation of one point in each connected component of a smooth real algebraic set. In *Proceedings of the 2003 International Symposium on Symbolic and Algebraic Computation (ISSAC '03)*, pages 224–231. ACM, 2003.

[31] Mohab Safey El Din and Pierre-Jean Spaenlehauer. Critical point computations on smooth varieties: Degree and complexity bounds. In *Proceedings of the 2016 ACM International Symposium on Symbolic and Algebraic Computation (ISSAC '16)*, pages 183–190. ACM, 2016.

[32] Sigurd Skogestad and Manfred Morari. Variable selection for decentralized control. *Modeling, Identification and Control*, 13(2):113–125, 1992.

[33] The authors. D-stability under first-order attachments at a single coordinate: an exact static threshold and a subset-sum criterion. Companion manuscript, 2026.

[34] The authors. Deciding D-stability is coNP-hard. Companion manuscript, 2026.